

Model Evaluation Report

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1 Introduction

This report presents a comparison of forecasting models trained on the gold price dataset from the US futures market. Two models were examined: a baseline and a long short term memory (LSTM) models.

The baseline approach focused on the single regression of the target price using the forecast of additional features. Specifically, ARIMA(1,1,1) was proposed to forecast high and low prices, and the XGBoost model was trained to forecast volume and close price.

Then, the LSTM model was trained based on the multivariate regression of relevant features. The two models were evaluated on the mean squared error and mean absolute error metrics. In addition, the models were tuned by a nature-inspired algorithm (particle swarm optimization).

The results suggest that although the LSTM model is unsuitable on a long time scale, the model produces reasonable results compared to the baseline model. The future analysis should investigate the approaches to further increase the metrics on the short-term predictions.

2 Method

The evaluation of models is performed on the US futures gold prices dataset with the hourly rate of the records. The data set captures prices from Summer 2023 and ends in the middle of June 2025.

The data set was split with 70% being train set and the rest being test set.

The following metrics were used: mean squared error and mean absolute error. The first metric captures significant errors of the predicted values, while the second metric identifies the absolute error values.

In addition, the predicted and true prices on the test set are plotted.

2.1 Baseline Model

The following ensemble architecture is used for the baseline model:

1. High price predictor (ARIMA)

2. Low price predictor (ARIMA)
3. Volume predictor (XGBoost regressor)
4. Close price predictor (XGBoost regressor)

2.2 LSTM Model

The following architecture of the LSTM model is proposed:

```
# self.lag = 24
# len(self.feature_names) = 14

model = Sequential([
    Input(shape=(len(self.feature_names), self.lag)),
    LSTM(128, return_sequences=True),
    Dropout(0.3),
    LSTM(64),
    Dense(32, activation='relu'),
    Dense(len(self.feature_names))
])
model.compile(optimizer='adam', loss='mse')
```

3 Results

The results of the comparison include the following metrics:

3.1 Metrics

Model	MSE	MAE
Baseline	2, 277, 247, 122	41, 213
LSTM	483, 948	642

Table 1: Metrics results

3.2 Plots

The predictions of prices compared with the true values are shown below in the figures [Fig. 1](#) and [Fig. 2](#). The blue curve represents the true values of the gold prices, while the orange line shows the predictions of each model.

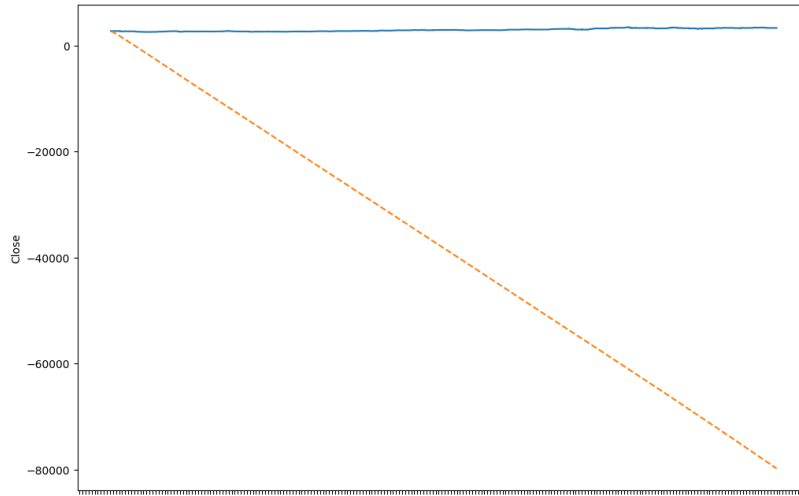


Figure 1: Baseline model predictions vs. true prices

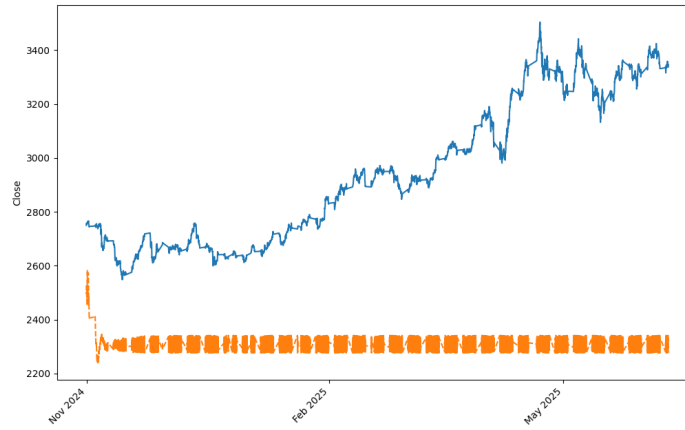


Figure 2: LSTM model predictions vs. true prices

4 Discussion

Clearly, both models are still far from accurate price predictions. Moreover, the baseline model demonstrated irrelevant results: the values went below zero. This behavior suggests more feature analysis. However, the LSTM model suc-

cessfully captured the general pattern during the first few weeks. Then, the model converged to some static behavior over time.

5 Conclusion

In conclusion, the LSTM model demonstrated better performance compared to the baseline model but still needs further tuning. The LSTM model approximates the timeseries behavior during the first several weeks but with a small bias. Therefore, the model can be used to predict the price rises and falls.